

35. Morphological Operations Using OpenCV – Black Hat Technique

Aim

To perform **morphological Black Hat transformation** on an image using OpenCV.

Procedure

1. Import the required OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a structuring element (kernel).
5. Apply the **Black Hat transformation** using OpenCV.
6. Display the original and processed images.
7. Save the output image.

Program

```
import cv2

import numpy as np

# Read the input image

img = cv2.imread("input.jpg")

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create structuring element

kernel = np.ones((5, 5), np.uint8)
```

```
# Apply Black Hat transformation

blackhat = cv2.morphologyEx(gray, cv2.MORPH_BLACKHAT, kernel)


# Display images

cv2.imshow("Original Image", img)

cv2.imshow("Black Hat Image", blackhat)


# Save output image

cv2.imwrite("blackhat_output.jpg", blackhat)

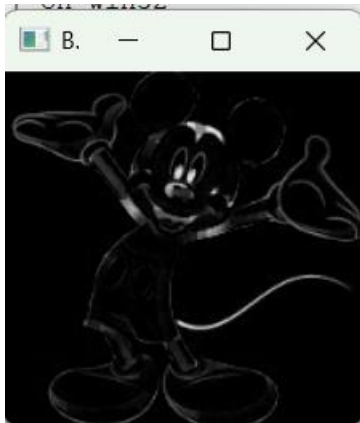

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the **morphological Black Hat operation** was successfully performed using OpenCV. It helps to **extract small dark objects and details from a bright background**.